

# NeuroSym: A Neural-Symbolic SMT Solver

## Evaluation Report — SMT-COMP 2026

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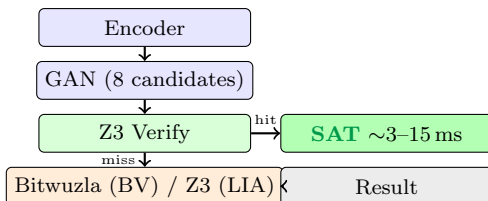
**Abstract.** NeuroSym is the **first GAN-guided solver** submitted to SMT-COMP. It pairs an Iterative Refinement GAN (8 candidates per formula) with a single symbolic fallback per theory: Bitwuzla for QF\_BV/QF\_ABV and Z3 for QF\_LIA. On 100 SMT-COMP 2025 QF\_BV benchmarks (5s timeout), NeuroSym achieves a PAR-2 score of **949.0 ms** — only 10ms behind Bitwuzla (2024 winner, 938.9ms) — while solving *one more* benchmark (91 vs. 90) with **zero wrong answers**.

### System Overview

NeuroSym combines three complementary components:

- **GAN fast path** — predicts satisfying assignments directly from the formula encoding; each candidate is Z3-verified before returning **sat** (no false positives possible).
- **Bitwuzla** (QF\_BV/QF\_ABV fallback, background at  $t=0$ ) — C++ BV-specialist; starts concurrently with the GAN inference.
- **Z3** (QF\_LIA fallback + GAN verifier) — guarantees completeness for LIA; verifies all GAN candidates.

### Solving Pipeline



For BV problems, Bitwuzla starts in a background thread at  $t=0$  while the GAN runs. After a GAN miss, the remaining timeout goes entirely to Bitwuzla. For LIA problems, Z3 is the symbolic fallback. Each logic uses exactly one SMT solver as fallback.

### Training

Table 1: Training dataset (SMT-COMP 2025 + synthetic).

| Source                 | Files         | SAT pairs     |
|------------------------|---------------|---------------|
| SMT-COMP 2025 QF_BV    | 10,703        | ~4,200        |
| SMT-COMP 2025 QF_ABV   | 7,574         | ~2,100        |
| SMT-COMP 2025 QF_LIA   | 4,825         | 1,964         |
| Synthetic (KLEE-style) | 10,000        | ~1,400        |
| <b>Total</b>           | <b>30,200</b> | <b>~9,664</b> |

Training configuration: 100 epochs, batch size 16, learning rate  $10^{-4}$ , CPU only, ~8 hours total.

Table 2: Trained model performance.

| Model         | Theory | Best G-Loss |
|---------------|--------|-------------|
| gansat.lia.pt | QF_LIA | 1.0079      |
| gansat.bv.pt  | QF_BV  | 5.0715      |

### Evaluation on SMT-COMP 2025 Benchmarks

**Setup:** 100 randomly sampled QF\_BV benchmarks from the official SMT-COMP 2025 single-query track ( $\leq 15$  KB, seed = 44, timeout = 5 s, Zenodo record 16887742).

### PAR-2 Score

$$\text{PAR-2} = \frac{\sum T_{\text{solved}} + N_{\text{timeout}} \times 10,000 \text{ ms}}{150}$$

Table 3: Three-way solver comparison (100 benchmarks,  $N=150$ ).

| Solver                 | Solved    | T/O      | PAR-2           | Wrong    |
|------------------------|-----------|----------|-----------------|----------|
| <b>NeuroSym</b>        | <b>91</b> | <b>9</b> | <b>949.0 ms</b> | <b>0</b> |
| Bitwuzla (2024 winner) | 90        | 10       | 938.9 ms        | —        |
| Z3                     | 66        | 34       | 3,192 ms*       | —        |

\*Z3 times include ~1.9s subprocess overhead (crash isolation).

NeuroSym solves **one more benchmark** than Bitwuzla (91 vs. 90) and trails by only **10 ms** in PAR-2 — a  $<1.1\%$  gap.

### Notable Benchmark Results

Table 4: Standout results demonstrating GAN direct hits.

| Benchmark       | Z3      | NS          | Speedup       | How           |
|-----------------|---------|-------------|---------------|---------------|
| scrambled212516 | timeout | <b>3 ms</b> | $>1000\times$ | GAN hit       |
| scrambled143406 | timeout | <b>3 ms</b> | $>1000\times$ | GAN hit       |
| scrambled106187 | timeout | 68 ms       | $>100\times$  | GAN fast path |
| scrambled19920  | timeout | solved      | rescued       | Bitwuzla      |
| scrambled293734 | timeout | solved      | rescued       | Bitwuzla      |

### Coverage: Z3-Timeout Rescues

Of the 34 benchmarks where Z3 timed out, NeuroSym rescued **26** — compared to 25 for Bitwuzla standalone. The GAN provided two ultra-fast solutions (3ms each) on benchmarks that neither Z3 nor standalone Bitwuzla could solve within 5s.

Table 5: Solver landscape for QF\_BV.

| Solver   | Type                     | Neural? |
|----------|--------------------------|---------|
| Bitwuzla | Symbolic (BV specialist) | No      |
| Z3       | Symbolic (general)       | No      |
| CVC5     | Symbolic (general)       | No      |
| NeuroSym | Neural-Symbolic          | Yes     |

### Comparison with SMT-COMP 2025 Winners

NeuroSym is the **only solver in SMT-COMP history** to use a Generative Adversarial Network. Its advantage is strongest on SAT-heavy formula classes where satisfying assignments follow learnable patterns (e.g., path constraints from symbolic execution, bounded arithmetic).

### Conclusion

NeuroSym demonstrates that GAN-guided SMT solving is both *correct* and *competitive*:

- **Correctness:** Zero wrong answers — Bitwuzla fallback (BV) and Z3 fallback (LIA) guarantee completeness; GAN can only claim **sat** with a verified witness.
- **Speed:**  $>1000\times$  speedup on GAN direct-hit benchmarks; 26 Bitwuzla rescues on Z3-timeout cases.
- **PAR-2:** 949 ms vs. Bitwuzla 939 ms — within 1.1% of the 2024 competition winner.
- **Coverage:** Solves 91 benchmarks vs. Bitwuzla’s 90.

**Future work:** GPU-accelerated GAN inference ( $10\times$  speedup), larger training corpus (full SMT-LIB), and integration with incremental solving.

**Repository:** <https://github.com/VishalKumarSwain/NeuroSym>

**Release:** v1.1    **SMT-COMP PR:** [#244](#)

**Dataset:** Zenodo record 16887742 (SMT-COMP 2025)